

An Introduction to Projective Geometry

Neil Makur

September 2022

Contents

1	A Tale of Mordor	2
2	Drawing	3
2.1	2D Projections	3
2.2	2-Point Perspective	4
3	Projective Transformations	4
3.1	1 Dimension	4
3.2	$\mathbb{RP}^1$ Revisited	7
3.3	2 Dimensions	8
4	Homogeneous Coordinates	8
4.1	Points and Lines in $\mathbb{RP}^2$	8
4.2	Homogeneous Functions	9
4.3	Some Conic Examples	10
5	Mordor Revisited	11
6	Geometric Applications	12
6.1	Desargues' Theorem	12
6.2	Pappus' Theorem	13
7	Conics	13
7.1	Matrices and Homogeneous Coordinates	13
7.2	Conics as Matrices	14
7.3	Transforming Conics	14

1 A Tale of Mordor

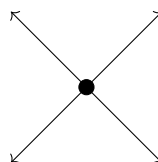
Two hobbits, Tomo and Jerrybo, go to Mordor disguised as orcs. They are both mostly fluent in orcish, and have to attend orc school. Tomo speaks fluent orcish, and Jerrybo is fluent except that he switches the words “point” and “line”.

Tomo and Jerrybo attend orc geometry class. Their teacher says that any two points determine a line. Tomo and Jerrybo both agree that this checks out:

Tomo

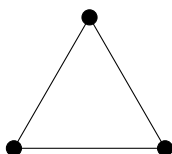


Jerrybo

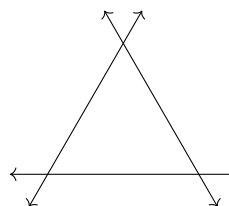


The teacher then says that any three non-collinear points determine a triangle. Once again, both Tomo and Jerrybo agree:

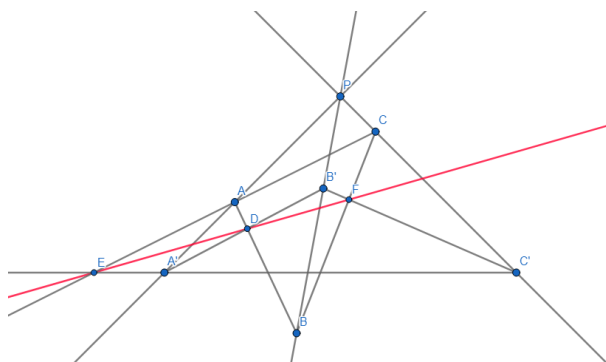
Tomo



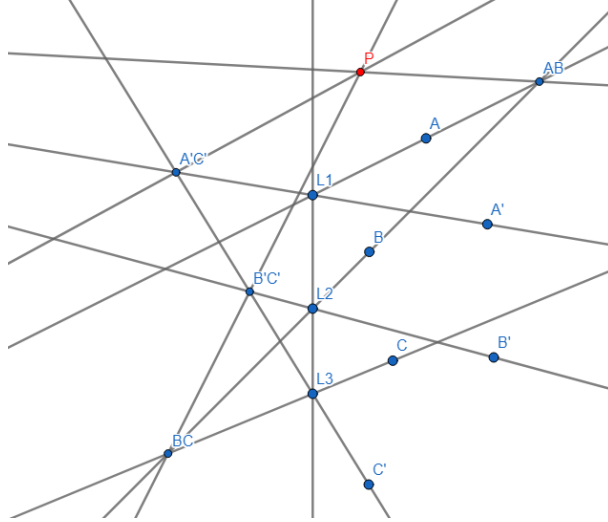
Jerrybo



The teacher then states Desargues' theorem: If ℓ_1, ℓ_2, ℓ_3 are lines all intersecting at a point, A, A' are points on ℓ_1 , B, B' are points on ℓ_2 , and C, C' are points on ℓ_3 , and $D = AB \cap A'B'$, $E = AC \cap A'C'$, $F = BC \cap B'C'$, then D, E, F are collinear. Tomo draws the following, and agrees:



Jerrybo draws the following and agrees:



This leads us to the following:

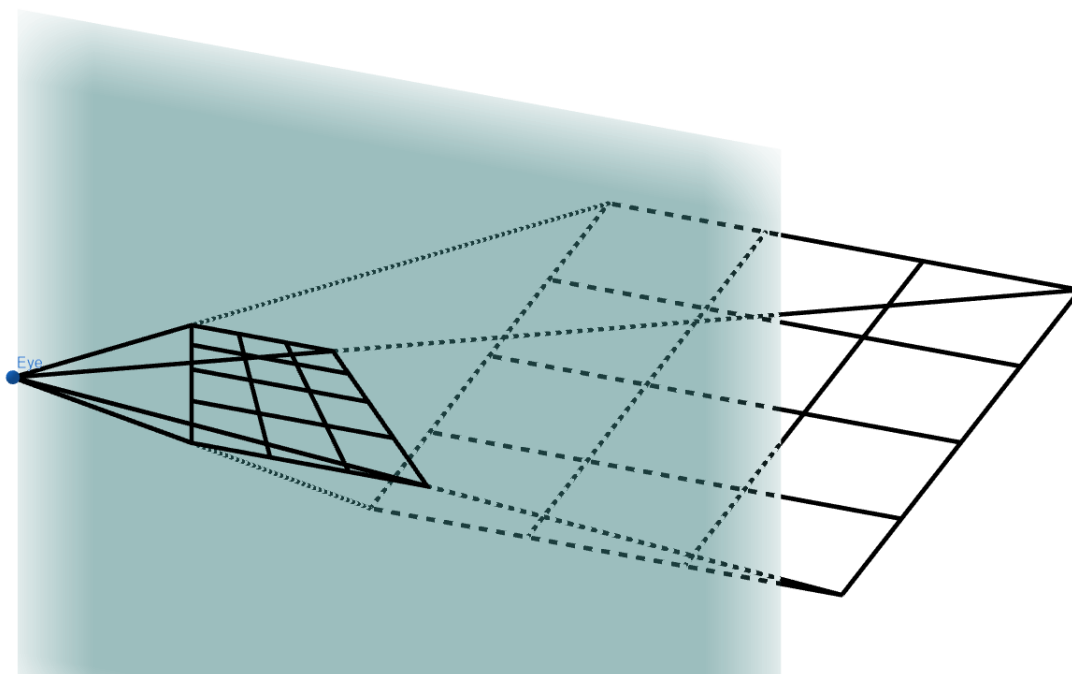
Conjecture 1.1 (Meta-Mordor Conjecture). *Given a true statement about points, lines, and their intersections that avoids the use of angles, length and parallel lines, we get another true statement if we swap the words “point” and “line”.*

There is some subtlety in the above statement. Firstly, we are not only allowed to talk about parallel lines, we are not allowed to *have* parallel lines at all. Swapping the words “point” and “line” in “Any two points give a line” gives “Any two lines give a point”. This is generally true, but not if the lines are parallel. Thus, we must somehow always be able to intersect lines. Secondly, we cannot literally swap the words “point” and “line”. For example, it makes no sense for three distinct points to be concurrent or for three distinct lines to be collinear. We will have to modify our understandings a little, understanding that concurrent points are collinear and that the intersection of two points is the line containing them, for example.

2 Drawing

2.1 2D Projections

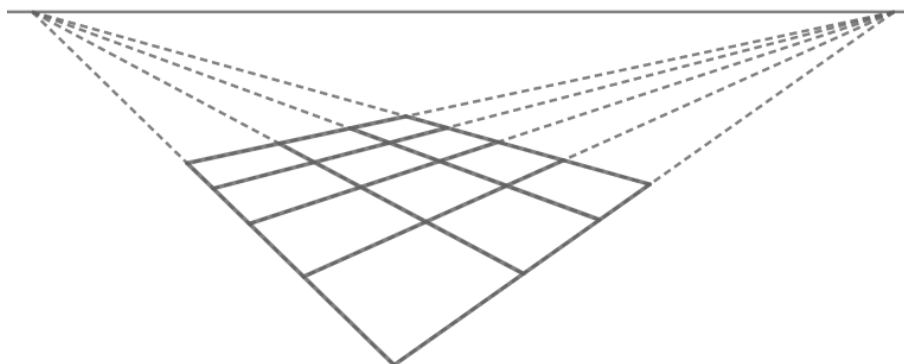
One popular method of drawing is to have an artist “project” what they are drawing onto their canvas. They imagine a line between their eye and the object they want to draw, and they draw it at the place where that line strikes the canvas.



We call maps like these “projective transformations”. We’ll call the point that we pass lines through “the Eye of Sauron”.

2.2 2-Point Perspective

Another popular method is to pick a “horizon line” where all pairs of parallel lines will intersect. Then, pick two points on that horizon line for the intersections of lines parallel to the two major axes. Drawing the checkerboard with this method, we get the following:



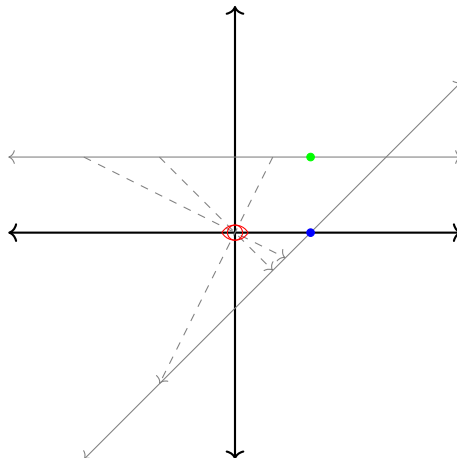
The drawing is actually determined by the choice of the horizon line, two points, and corners of the frontmost square. This is because the diagonals of the checkerboard are parallel and so must intersect at the same point on the horizon line.

3 Projective Transformations

3.1 1 Dimension

Let us look at some examples of projective transformations in 1 dimension. For the sake of simplicity, we place the Eye of Sauron at the origin. Suppose that we are projecting the line $y = 1$ onto the line $y = x - 1$,

as below:

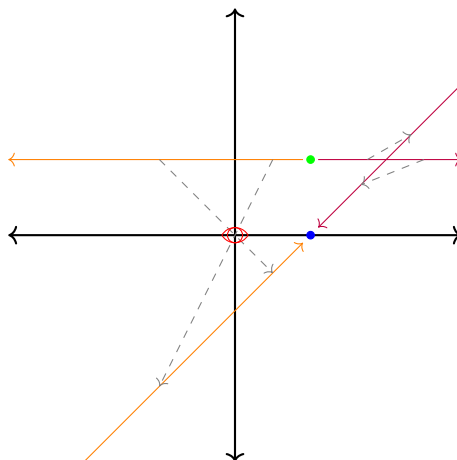


Call $y = 1$ the “subject line”, and $y = x - 1$ the “canvas line” (so that we are “drawing” the subject on the canvas). Let us try to find an algebraic way to describe this transformation. We can parameterize $y = 1$ by $(t, 1)$ as t ranges over $\mathbb{R}$ and $y = x - 1$ by $(s, s - 1)$ as s ranges over $\mathbb{R}$. We can now describe our projective transformation by describing how to obtain s from t when $(t, 1) \mapsto (s, s - 1)$.

The line between $(t, 1)$ and $(0, 0)$ is $y = \frac{1}{t}x$. We want to find the intersection with $y = x - 1$, which happens when $\frac{1}{t}x = x - 1$. Solving, $x = \frac{t}{t-1}$. Thus, $s = \frac{t}{t-1}$. Hence, our map is described by

$$t \mapsto \frac{t}{t-1}.$$

However, there are some problems with this map. Firstly, the point $(1, 1)$ (marked in green), which corresponds to $t = 1$, does not map to any point under our transformation. As we get closer and closer to the green point, the function maps to points further and further along the canvas line. Secondly, the point $(1, 0)$ (marked in blue), which corresponds to $s = 1$, is not mapped to by any point on the subject line, but as we go further and further along the subject line, we get closer and closer to the blue point. This is illustrated below.



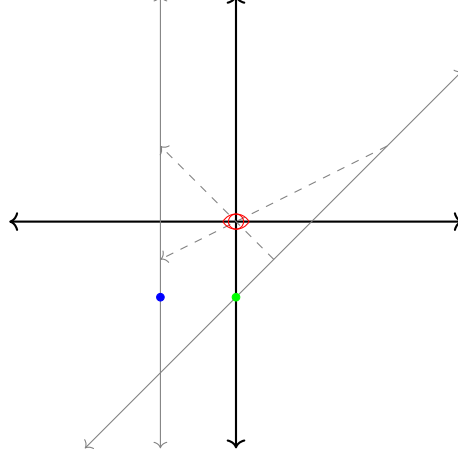
Taking inspiration from the horizon line in 2-point perspective drawing, we create a “point at infinity” in both our subject and canvas planes. We can parameterize our subject plane as $(t, 1)$, now allowing t to take values in $\mathbb{R} \cup \{\infty\}$, and our canvas plane as $(s, s - 1)$ with s taking values in $\mathbb{R} \cup \{\infty\}$. Our map now becomes

$$t \mapsto \begin{cases} \frac{t}{t-1} & t \neq 1, \infty \\ \infty & t = 1 \\ 1 & t = \infty \end{cases},$$

which is a map from $\mathbb{R} \cup \{\infty\}$ to itself.

Definition 3.1. The *Real Projective Line* is $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$.

Armed with our knowledge about $\mathbb{RP}^1$, let us tackle another example. Suppose that our subject line is $y = x - 1$ and that our canvas line is $x = -1$, as below.



Our special points are once again marked.

Now, we can parameterize our subject line as $(s, s - 1)$ for $s \in \mathbb{RP}^1$ and our canvas line as $(-1, r)$ for $r \in \mathbb{RP}^1$. Our green point occurs at $s = 0$ and our blue point occurs at $r = -1$. When $s \neq 0, \infty$, the line through $(0, 0)$ and $(s, s - 1)$ is $y = \frac{s-1}{s}x$. This intersects $x = -1$ at $(-1, -\frac{s-1}{s})$. Hence, for $s \neq 0, \infty$, our map is $s \mapsto \frac{1-s}{s}$. Using the values of s and r at our green and blue points respectively, we get the following map

$$s \mapsto \begin{cases} \frac{1-s}{s} & s \neq 0, \infty \\ \infty & s = 0 \\ -1 & s = \infty \end{cases}.$$

Now, we've found a projective transformation from $y = 1$ to $y = x - 1$ and $y = x - 1$ to $x = -1$. We would expect that composing these would give a projective transformation from $y = 1$ to $x = -1$. Let us check this. Our two maps are

$$p_1 : t \mapsto \begin{cases} \frac{t}{t-1} & t \neq 1, \infty \\ \infty & t = 1 \\ 1 & t = \infty \end{cases} \quad p_2 : s \mapsto \begin{cases} \frac{1-s}{s} & s \neq 0, \infty \\ \infty & s = 0 \\ -1 & s = \infty \end{cases}.$$

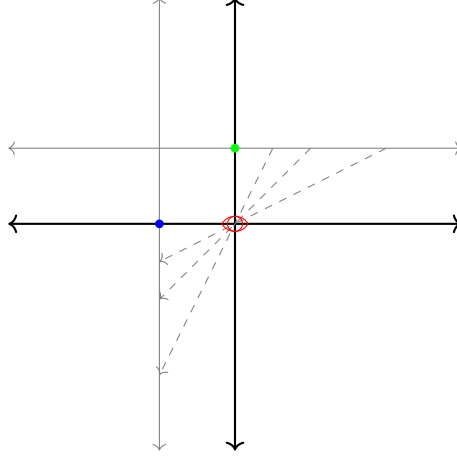
Our values of consideration in the first map are $t = 1, \infty$. Our values of consideration in the second map are $s = 0, \infty$, which occur at $t = 0, 1$ respectively. Hence, we set special consideration for the values of $t = 0, 1, \infty$. If $t \neq 0, 1, \infty$, we have that

$$(p_2 \circ p_1)(t) = \frac{1 - \frac{t}{t-1}}{\frac{t}{t-1}} = -\frac{1}{t}.$$

If $t = 0$, $p_1(t) = 0$, so $(p_2 \circ p_1)(t) = \infty$. If $t = \infty$, $p_1(t) = 1$, so $(p_2 \circ p_1)(t) = 0$. Finally, if $t = 1$, $p_1(t) = \infty$, so $(p_2 \circ p_1)(t) = -1$, which is conveniently $-\frac{1}{t}$. Thus,

$$p_2 \circ p_1 : \begin{cases} -\frac{1}{t} & t \neq 0, \infty \\ \infty & t = 0 \\ 0 & t = \infty \end{cases}.$$

Let us compute the regular projection from $y = 1$ to $x = -1$. We have the following:



Our green and blue points occur at $t = 0$ and $r = 0$ respectively. When $t \neq 0, \infty$, the line through $(t, 1)$ is $y = \frac{1}{t}x$, which intersects $x = -1$ at $(-1, -\frac{1}{t})$, so that $t \mapsto -\frac{1}{t}$. Hence, our map is

$$t \mapsto \begin{cases} -\frac{1}{t} & t \neq 0, \infty \\ \infty & t = 0 \\ 0 & t = \infty \end{cases},$$

which matches our composition. Notice that, if we only worked over $\mathbb{R}$ and not $\mathbb{RP}^1$, we would not have been able to figure out where $t = 1$ maps in the composition, even though there is no problem with this in the actual projection between $y = 1$ and $x = -1$.

3.2 $\mathbb{RP}^1$ Revisited

Our current definition of $\mathbb{RP}^1$ is the set $\mathbb{R} \cup \{\infty\}$. However, this is sort of clunky: we have to handle ∞ as a separate case from the rest of $\mathbb{R}$. Let's try to find a way to avoid this.

Consider the copy of $\mathbb{RP}^1$ that we have on the line $y = 1$. We have the line itself along with its point at infinity. Consider what happens when we project onto the line $x = -1$. We draw lines through each point on $y = 1$, giving us all the lines through $(0, 0)$ except for $y = 0$. The intersection of $y = 0$ and $x = -1$ is mapped to by the point at infinity. Thus, just as the line $y = \frac{1}{t}x$ corresponds to the point $(t, 1)$, the line $y = 0$ corresponds to the point at infinity. It therefore seems that looking at lines through the origin is more canonical than looking at a line with a point at infinity. This prompts the following:

Definition 3.2. The Real Projective Line is the set

$$\mathbb{RP}^1 = \{\text{lines in } \mathbb{R}^2 \text{ passing through } \odot\}.$$

We can parameterize the elements of $\mathbb{RP}^1$ as (at, bt) for some fixed $a, b \in \mathbb{R}$ not both 0 and varying $t \in \mathbb{R}$. However, this is not unique: for example, (t, t) and $(2t, 2t)$ are the same line. To fix this, we can “homogenize” the lines by seeing where they intersect $y = 1$. If $b \neq 0$, we get that $x = \frac{a}{b}$. If $b = 0$, we get the point at infinity. This gives us back our old definition of $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$. Using these ideas, let us up the dimension.

Definition 3.3. Real Projective Space is the set

$$\mathbb{RP}^n = \{\text{lines in } \mathbb{R}^{n+1} \text{ passing through } \odot\}.$$

We can once again parameterize the elements of $\mathbb{RP}^n$ as $(a_1t, a_2t, \dots, a_{n+1}t)$ for fixed $a_1, \dots, a_{n+1} \in \mathbb{R}$ not all 0 and varying $t \in \mathbb{R}$. When $n = 0$, there is only one element, which we will denote ∞ , so that $\mathbb{RP}^0 = \{\infty\}$. When $n \geq 1$, we once again run into the uniqueness issue. We can intersect the lines with the plane $x_{n+1} = 1$. If $a_{n+1} \neq 0$, we get the unique point $(a_1/a_{n+1}, a_2/a_{n+1}, \dots, a_n/a_{n+1}, 1)$. If $a_{n+1} = 0$,

the line never intersects $x_{n+1} = 1$, so we have coordinates of the form $(a_1t, a_2t, \dots, a_nt, 0)$, with the same uniqueness issue. Ignoring the last coordinate, this is exactly $\mathbb{RP}^{n-1}$. Hence, we have derived the following:

$$\mathbb{RP}^n = \mathbb{R}^n \cup \mathbb{RP}^{n-1}.$$

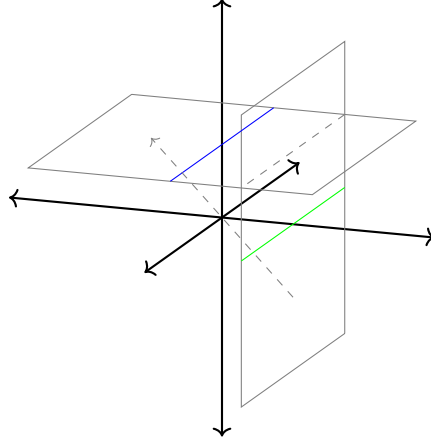
Here, $\mathbb{RP}^{n-1}$ contains what is “at infinity”. For example, in the Real Projective Plane $\mathbb{RP}^2$, the $\mathbb{RP}^1$ is the line at infinity, and in the Real Projective Line $\mathbb{RP}^1$, the $\mathbb{RP}^0$ is the point at infinity. Recursively, we get that

$$\mathbb{RP}^n = \mathbb{R}^n \cup \mathbb{R}^{n-1} \cup \dots \cup \mathbb{R}^2 \cup \mathbb{R}^1 \cup \{\infty\}.$$

The process to get this equivalence is to homogenize the rightmost nonzero coordinate to 1, and only look at everything to the left of that (this is just repetition of intersecting with the plane $x_{n+1} = 1$). This gives us a unique way to represent elements of $\mathbb{RP}^n$.

3.3 2 Dimensions

We will project the plane $y = 1$ onto the plane $z = 1$.



We first discuss how we will represent the lines at infinity of the two planes. The points of $\mathbb{RP}^2$ that are on the line at infinity of the subject plane are those of the form $(at, 0, bt)$. We intersect these with the line given by $z = 1$ and $y = 0$. If the x -coordinate of this intersection is x , we call this line $\ell_\infty^s(x)$. If the line does not intersect the line $z = 1$ and $y = 0$, then we call the line ∞ . The points of $\mathbb{RP}^2$ that are on the line at infinity of the canvas plane are those of the form $(at, bt, 0)$. We intersect these with the line given by $y = 1$ and $z = 0$. If the x -coordinate of this intersection is x , we call this line $\ell_\infty^c(x)$. If the line does not intersect the line $y = 1$ and $z = 0$, we call the line ∞ (note that these are the same line, so there is no ambiguity).

With this notation, we can describe our transformation. Firstly, if $(s, 1, t)$ is a point on the subject plane that is not on the green line, then it will be mapped to $(s/t, 1/t, 1)$. If $t = 0$, the point is mapped to $\ell_\infty^c(s)$. Now, $\ell_\infty^s(x)$ intersects the line $z = 1$ and $y = 0$ at $(x, 0, 1)$, and so this is the point to which it maps. Finally, ∞ maps to ∞ , since it never intersects either plane. Hence, our map is the following

$$\begin{cases} (s/t, 1/t, 1) & (s, 1, t), t \neq 0 \\ \ell_\infty^c(s) & (s, 1, 0) \\ (x, 0, 1) & \ell_\infty^s(x) \\ \infty & \infty \end{cases}.$$

4 Homogeneous Coordinates

4.1 Points and Lines in $\mathbb{RP}^2$

We know that the “points” in $\mathbb{RP}^2$ are the lines through $\odot$ in $\mathbb{R}^3$. A “line” in $\mathbb{RP}^2$ should be something that is uniquely determined by two distinct points in $\mathbb{RP}^2$. It therefore makes sense to say that the “lines” in $\mathbb{RP}^2$ are the planes in $\mathbb{R}^3$ passing through $\odot$.

To represent the “point” $(at, bt, ct) \in \mathbb{RP}^2$, we will write $(a : b : c)$. To represent the “line” $ax + by + cz = 0$, we will write $[a, b, c]$. We call these homogeneous coordinates. In both of these, at least one of a, b, c is nonzero. We will more generally write $(x_1 : x_2 : \dots : x_{n+1})$ for the “point” $(x_1t, x_2t, \dots, x_{n+1}t) \in \mathbb{RP}^n$.

Let us try to “dehomogenize” our coordinates with a subject plane of $z = 1$. The “point” $(x : y : z)$ intersects $z = 1$ at $(x/z, y/z, 1)$, unless $z = 0$, in which case we have a “point” at infinity. That is, the “points” at infinity are of the form $(x : y : 0)$. The “line” $[a, b, c]$ intersects the subject plane at the line given by $ax + by + c = 0$ and $z = 1$, which is the line $ax + by + c = 0$ on the subject plane. However, if $a = b = 0$, this “line” does not intersect the plane $z = 1$ anywhere. The “line” is $[0, 0, c] = [0, 0, 1]$, which corresponds to the plane $z = 0$. This is the “line” at infinity.

Suppose that we have two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ for $c_1 \neq c_2$ in the subject plane. These correspond to the “lines” $[a, b, c_1]$ and $[a, b, c_2]$, which are the planes $ax + by + c_1z = 0$ and $ax + by + c_2z = 0$. In the intersection of these planes, we must have $(c_2 - c_1)z = 0$ so that $z = 0$. Plugging this in, we must also have $ax + by = 0$. Thus, the intersection of these planes is the line parameterized by $(-bt, at, 0)$, which is the “point” $(-b : a : 0)$. We know that this “point” is a “point” at infinity. Hence, two “lines” that seem parallel in the subject plane actually intersect at a “point” at infinity.

Let us turn to the problem of checking if a “point” is on a “line”. The “point” $(x_0 : y_0 : z_0)$ corresponds to the line (x_0t, y_0t, z_0t) , and the “line” $[a, b, c]$ corresponds to the plane $ax + by + cz = 0$. Thus, the “point” $(x_0 : y_0 : z_0)$ lies on the “line” $[a, b, c]$ if and only if $ax_0t + by_0t + cz_0t = 0$ for all $t \in \mathbb{R}$. Thus, $(x_0 : y_0 : z_0)$ lies on $[a, b, c]$ if and only if $ax_0 + by_0 + cz_0 = 0$. If we write the “point” P as a column vector and the “line” L as a row vector, this condition becomes $LP = 0$.

4.2 Homogeneous Functions

Definition 4.1. A homogeneous function is a function f with the property that

$$f(sx_1, sx_2, \dots, sx_n) = s^k f(x_1, x_2, \dots, x_n)$$

for some integer k .

For example, $f(x) = x^2$ is homogeneous with $k = 2$, since $f(sx) = s^2x^2 = s^2f(x)$. However, $f(x) = x^2 + x + 1$ is not homogeneous, since $f(sx) = s^2x^2 + sx + 1$, which we cannot write as $s^k f(x)$ for some integer k .

If we start with a point $(x_1, \dots, x_n) \in \mathbb{R}^n$, we can homogenize the coordinates and bring them into $\mathbb{RP}^n$ by multiplying by a nonzero Z to get $(x_1Z : \dots : x_nZ : Z)$. If $Z = 1$, we have $(x_1, \dots, x_n, 1)$ in the $Z = 1$ subject plane. If we write $x_i = \frac{X_i}{Z}$, we get that this is $(X_1 : X_2 : \dots : X_n : Z)$.

Why do we call these homogeneous coordinates? If we have a polynomial function $f(x_1, \dots, x_n)$ of degree d , then the polynomial function given by

$$F(X_1, \dots, X_n, Z) = Z^d f(X_1/Z, \dots, X_n/Z)$$

is homogeneous. This is because, writing

$$f(x_1, \dots, x_n) = \sum c_i x_1^{a_{1,i}} x_2^{a_{2,i}} \dots x_n^{a_{n,i}},$$

we get that

$$F(X_1, \dots, X_n, Z) = \sum c_i X_1^{a_{1,i}} X_2^{a_{2,i}} \dots X_n^{a_{n,i}} Z^{d-a_{1,i}-\dots-a_{n,i}},$$

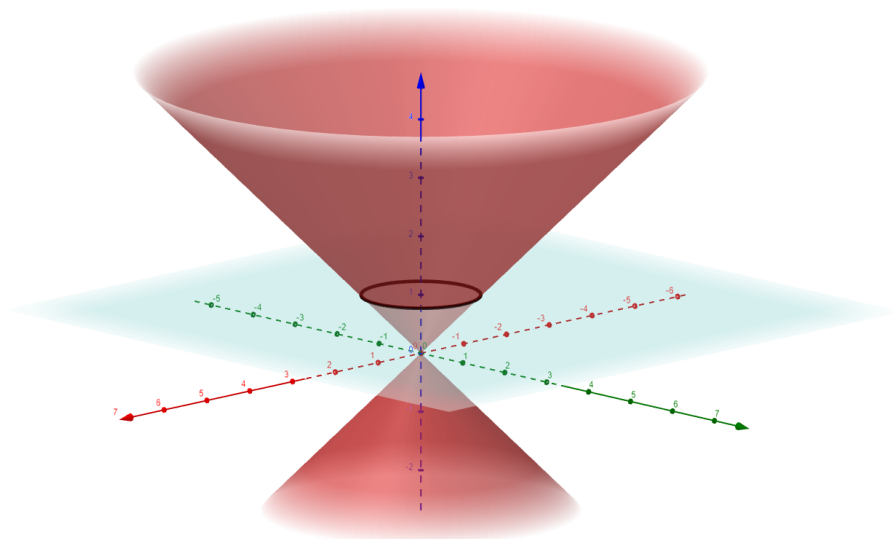
which is homogeneous with $k = d$.

At first glance, this still seems useless: if we have two ways of representing a “point” in $\mathbb{RP}^n$, $(X_1 : \dots : X_n : Z)$ and $(X'_1 : \dots : X'_n : Z)$, we are not guaranteed that F takes the same values on these. However, whether or not F is zero is preserved with these two representations of the same “point”. Thus, instead of trying to use these coordinates to take *polynomial functions* from $\mathbb{R}^n$ to $\mathbb{RP}^n$, we can use these coordinates to take *polynomial curves* from $\mathbb{R}^n$ to $\mathbb{RP}^n$. If we have a polynomial curve described by $f(x_1, \dots, x_n) = 0$ in $\mathbb{R}^n$, we can get a polynomial curve described by $F(X_1, \dots, X_n, Z) = 0$ in $\mathbb{RP}^n$ for a homogeneous polynomial F . Since F is homogeneous, this is actually a well-defined curve. This intersects the $Z = 1$ subject plane at points with $f(X_1, \dots, X_n) = 0$.

For example, consider the line $ax + by + c = 0$ in $\mathbb{R}^2$. Homogenizing, this is $aX + bY + cZ = 0$, which is the “line” $[a, b, c]$. Thus, this process of homogenizing a line gives exactly what we would expect it to give.

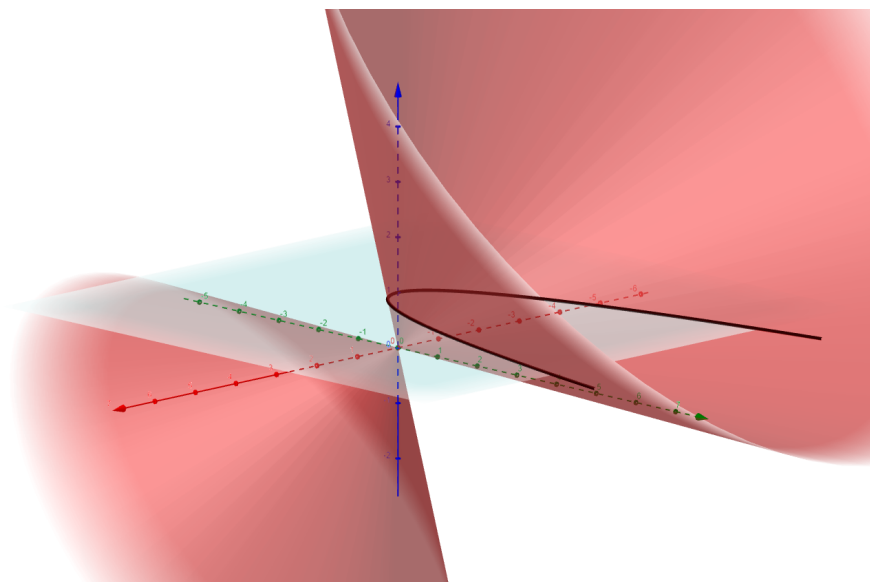
4.3 Some Conic Examples

Homogenizing the circle $x^2 + y^2 - 1 = 0$ gives $X^2 + Y^2 - Z^2 = 0$. Graphing this, we get the following:



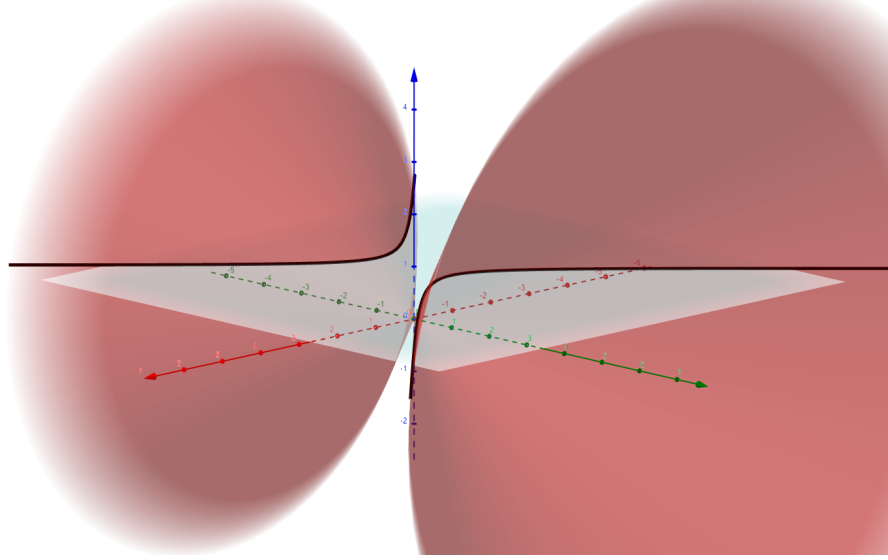
This seems to be a quite natural extension of the circle in $\mathbb{R}^2$ into $\mathbb{RP}^2$. However, extensions are not always this natural: we sometimes get sneaky “points” at infinity.

Consider what happens when we homogenize the parabola $y - x^2 = 0$ to get $YZ - X^2 = 0$. The graph of this is below:



This “parabola” has the lines passing through $\odot$ and a point on the parabola that it would expect it to have. However, it also has one more line. When $Z = 0$, we can have Y take on any value as long as $X = 0$. Thus, the line $X = Z = 0$ (or the “point” $(0 : 1 : 0)$) is also included. Therefore, this “parabola” in $\mathbb{RP}^2$ also contains one “point” at infinity. Alternatively, the “parabola” intersects the “line” at infinity at one “point”.

What if we homogenize the hyperbola $y^2 - x^2 - 1 = 0$ to get $Y^2 - X^2 - Z^2 = 0$? We get the following:



For this to intersect the “line” at infinity, we must have $Z = 0$, so that $(Y + X)(Y - X) = 0$. Thus, either $Y = X$ or $Y = -X$. Hence, the “hyperbola” in $\mathbb{RP}^2$ contains two “points” at infinity: $(1 : 1 : 0)$ and $(1 : -1 : 0)$.

5 Mordor Revisited

Recall that, in the Meta-Mordor conjecture, we can intersect any two lines. Although this is not generally true in $\mathbb{R}^2$, it *is* true in $\mathbb{RP}^2$!

In $\mathbb{RP}^2$, a “point” is a line through $\odot$ in $\mathbb{R}^3$, and a “line” is a plane through $\odot$ in $\mathbb{R}^3$. Now, the type of statements that the Meta-Mordor conjecture applies to are built out of two axioms: any two distinct “lines” must intersect in a “point”, and any two distinct “points” determine a “line”. In addition, it uses the processes of checking if a “point” is on a “line” and checking if a “line” contains a “point”.

Let us show that the first axiom is true. For two “lines” $[a, b, c]$ and $[d, e, f]$ to intersect in a “point” $(x : y : z)$, we must have that $ax + by + cz = 0$ and $dx + ey + fz = 0$. We can put this into a matrix $\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$ and row-reduce. We will obtain one free variable, and will be able to express the other two as a scalar multiple of this free variable. Since (nonzero) multiples in homogenized coordinates of “points” do not affect what the “point” is, we can set this free variable to 1 to find the intersection.

Now let us show that the second axiom is true. For two “points” $(a : b : c)$ and $(d : e : f)$ to lie on a “line” $[x, y, z]$, we must have that $xa + yb + zc = 0$ and $xd + ye + zf = 0$. We can put this into a matrix $\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$ and row-reduce. We will obtain one free variable, and will be able to express the other two as a scalar multiple of this free variable. Since (nonzero) multiples in homogenized coordinates of “lines” do not affect what the “line” is, we can set this free variable to 1 to find the line.

Now, if the “point” $(x : y : z)$ is on the “line” $[a, b, c]$, then $ax + by + cz = 0$. This is the exact same as saying that the “line” $[x, y, z]$ contains the “point” $(a : b : c)$. Similarly, saying that the “line” $[a, b, c]$ contains the “point” $(x : y : z)$ is the same as saying that the “point” $(a : b : c)$ lies on the “line” $[x, y, z]$.

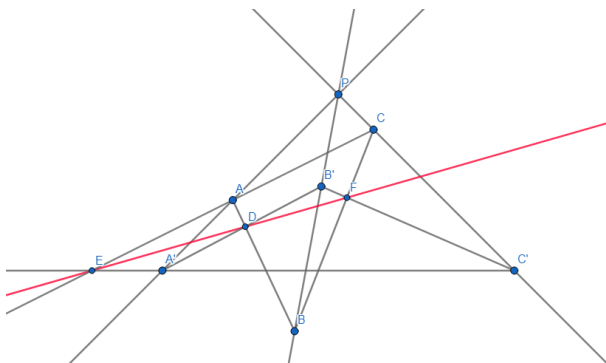
In the above paragraphs, the algebra used is the exact same. Thus, the process of intersecting two “lines” turns into the process of drawing a “line” through two “points” when we swap the coordinates of “points” and “lines” (i.e. replace $(x : y : z)$ with $[x, y, z]$ and $[a, b, c]$ with $(a : b : c)$), and vice versa. Further, the process of checking whether a “point” is on a “line” turns into the process of checking whether a “line” contains a “point”, and vice versa. Thus, whenever we build a statement out of intersecting “lines”, drawing “lines” through “points”, checking if “lines” contain “points”, and checking whether “points” lie on “lines”, the algebra stays the same when we swap the coordinates of “points” and “lines”, and so the validity of such a statement is not affected.

There is sometimes a third axiom that we consider: there exist 4 “points”, no 3 of which are on a “line”. This axiom implies itself under swapping “points” and “lines”. If the 4 points are A, B, C, D , then the lines AB, BC, CD, DA give the axiom after swapping “points” and “lines”. Conversely, if we assume the swapped axiom, and have 4 lines L_1, L_2, L_3, L_4 , then the points $L_1 \cap L_2, L_2 \cap L_3, L_3 \cap L_4, L_4 \cap L_1$ give the original axiom.

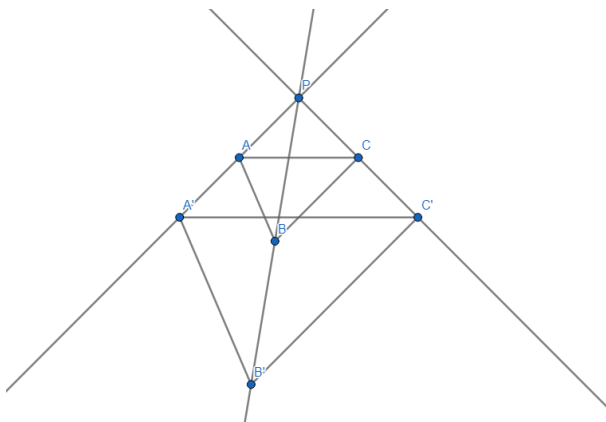
6 Geometric Applications

6.1 Desargues’ Theorem

Recall Desargues’ Theorem: If ℓ_1, ℓ_2, ℓ_3 are lines all intersecting at a point, A, A' are points on ℓ_1 , B, B' are points on ℓ_2 , and C, C' are points on ℓ_3 , and $D = AB \cap A'B'$, $E = AC \cap A'C'$, $F = BC \cap B'C'$, then D, E, F are collinear.



We will prove this with projective geometry. We’ll imagine this figure on the $z = 1$ plane. If we were to project to a new plane, we will have a new figure with intersections preserved. With our projective geometry, we can pick a line in our figure to send to the line at infinity in our new plane. We’ll send the line DF to infinity, and then look at E . Sending DF to infinity means that we send both D and F to infinity. Thus, AB and $A'B'$ are parallel, and BC and $B'C'$ are parallel.

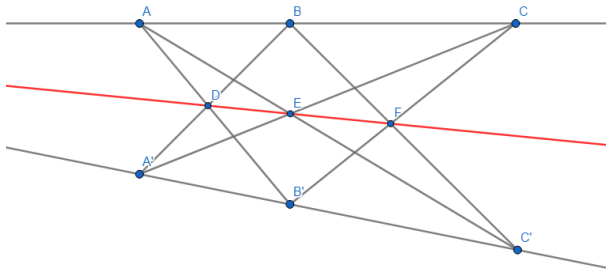


From the diagram alone, it is clear that AC and $A'C'$ are parallel. We can also prove this by realizing that the quadrilaterals $PCBA$ and $PC'B'A'$ are similar, so the triangles PCA and $PC'A'$ are similar, meaning that AC is parallel to $A'C'$. This means that we also sent E to the line at infinity of the new plane, so that D, E, F are collinear.

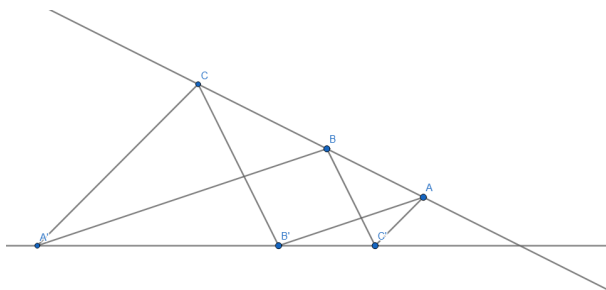
A slick proof of Desargues’ Theorem is to imagine ℓ_1, ℓ_2, ℓ_3 as forming a tripod in 3D, so that $\triangle ABC$ and $\triangle A'B'C'$ cut two planes in this 3D space. Then, as long as these planes aren’t parallel, they intersect at a line. If they are parallel, D, E, F cannot exist.

6.2 Pappus' Theorem

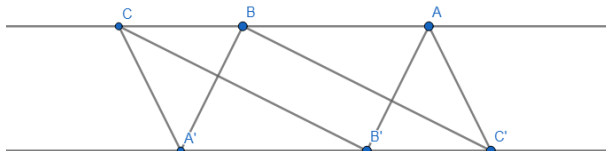
Pappus' Theorem states the following: If L and M are distinct lines with $A, B, C \in L$ and $A', B', C' \in M$, $D = AB' \cap A'B$, $E = AC' \cap A'C$, $F = BC' \cap B'C$, then D, E, F are collinear.



We can use the same trick of sending DF to the line at infinity and seeing where E ends up. Here, we have two cases. Either L and M are not parallel in our new plane, in which case we have something like



or L and M are parallel in our new plane, in which case we have something like



In both cases, we can use elementary geometry to prove our result.

7 Conics

7.1 Matrices and Homogeneous Coordinates

Suppose that we have a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ represented by a 2×2 matrix A . Then, we can use the matrix $\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$ to represent the action of A in $\mathbb{RP}^2$. If $A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$, the action of this matrix on the “point” $(x : y : 1)$ (written as a column vector) is $(x' : y' : 1)$. Thus, if we place $\mathbb{R}^2$ in the $z = 1$ subject plane, we can use a matrix to replicate the action of A on $\mathbb{R}^2$ in $\mathbb{RP}^2$.

However, linear transformations are not the only transformations that we can get from matrices in $\mathbb{RP}^2$. Consider, for example, the matrix

$$\begin{pmatrix} 1 & 0 & v_1 \\ 0 & 1 & v_2 \\ 0 & 0 & 1 \end{pmatrix}.$$

If we act this on the “point” $(x : y : 1)$, we get the “point” $(x + v_1 : y + v_2 : 1)$. Hence, we can also represent translations in $\mathbb{R}^2$ with matrices!

What happens if we were to apply a 3×3 matrix A to all the “points” on a “line”? Let us write the “line” $[a, b, c]$ as a row vector L , and the “point” $(x : y : z)$ as a column vector P . Then, if $LP = 0$, we consider the point AP . Therefore, if A is invertible, we can consider the “line” corresponding to LA^{-1} . This is because $(LA^{-1})(AP) = LP$, so $LP = 0$ (meaning we consider the point AP) if and only if $(LA^{-1})(AP) = 0$ (meaning that AP lies on the “line” LA^{-1}). What if A is not invertible? This means that we have some $(x, y, z) \in \mathbb{R}^3 \setminus \{\text{red circle}\}$ that maps to $(0, 0, 0)$ under A . Then, the “point” (x, y, z) will map to $(0 : 0 : 0)$. However, the “point” $(0 : 0 : 0)$ is not a part of $\mathbb{RP}^2$. Thus, non-invertible matrices are not maps from $\mathbb{RP}^2$ to $\mathbb{RP}^2$, and we do not have to consider them.

7.2 Conics as Matrices

Suppose we have a conic $ax^2 + bxy + cy^2 + dx + ey + f = 0$. We will represent this with the matrix equation

$$\begin{pmatrix} x & y & 1 \end{pmatrix} \begin{pmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0.$$

Note that this matrix is not unique, as we can scale it by any nonzero constant and get the same conic. Here are matrices for $x^2 + y^2 = 1$, $y = x^2$, and $y^2 - x^2 = 1$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

7.3 Transforming Conics

What happens if we were to apply a matrix A to all points on this conic? That is, if

$$\begin{pmatrix} x & y & 1 \end{pmatrix} \begin{pmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = 0,$$

we consider the point $A \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$. Once again, A must be invertible to make sense. We can write

$$0 = \left(A^{-1} A \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \right)^T \begin{pmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{pmatrix} \left(A^{-1} A \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \right) = \left(A \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \right)^T (A^{-1})^T \begin{pmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{pmatrix} (A^{-1}) \left(A \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \right).$$

Thus, we get a conic corresponding to the matrix

$$(A^{-1})^T \begin{pmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{pmatrix} (A^{-1}).$$

Using this fact, let us see what happens if we rotate a conic by an angle of $-\theta$. We would have

$$A^{-1} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

to be the matrix corresponding to rotation by θ . Applying this to our conic, we get the matrix

$$\begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is

$$\begin{pmatrix} a \cos^2 \theta + b \sin \theta \cos \theta + c \sin^2 \theta & -a \sin \theta \cos \theta + \frac{b}{2}(\cos^2 \theta - \sin^2 \theta) + c \sin \theta \cos \theta & \frac{d}{2} \cos \theta + \frac{e}{2} \sin \theta \\ -a \sin \theta \cos \theta + \frac{b}{2}(\cos^2 \theta - \sin^2 \theta) + c \sin \theta \cos \theta & a \sin^2 \theta - b \sin \theta \cos \theta + c \cos^2 \theta & -\frac{d}{2} \sin \theta + \frac{e}{2} \cos \theta \\ \frac{d}{2} \cos \theta + \frac{e}{2} \sin \theta & -\frac{d}{2} \sin \theta + \frac{e}{2} \cos \theta & f \end{pmatrix}.$$

Notice that this does correspond to a conic matrix, as it is its own transpose. Let us make the b of this new conic zero, to get a conic that is easy to draw from its equation. We have

$$0 = -a \sin \theta \cos \theta + \frac{b}{2}(\cos^2 \theta - \sin^2 \theta) + c \sin \theta \cos \theta = \frac{c-a}{2} \sin 2\theta + \frac{b}{2} \cos 2\theta.$$

Solving, we get that

$$\theta = \frac{1}{2} \arctan\left(\frac{b}{a-c}\right),$$

where we would define $\arctan(\infty) = \frac{\pi}{2}$ to deal with the case of $a-c=0$. Thus, we have shown that every conic can be rotated to a new conic with $b=0$. In other words, we can rotate any conic to align with the axes.

Notice that

$$\begin{pmatrix} X & Y \\ Z & w \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & c \end{pmatrix} = \begin{pmatrix} XA & cY \\ ZA & cw \end{pmatrix} \quad \begin{pmatrix} A & 0 \\ 0 & c \end{pmatrix} \begin{pmatrix} X & Y \\ Z & w \end{pmatrix} = \begin{pmatrix} AX & AY \\ cZ & cw \end{pmatrix},$$

where X and A are 2×2 matrices, c and w are numbers, and all other values have the appropriate dimensions. In particular, the upper left 2×2 matrices are XA and AX respectively. Thus, if $\det A = 1$, the determinants of the upper left 2×2 matrices are both $\det X$. Since this is true of our two matrices that we multiply by in a rotation, we get that the determinant of the upper left 2×2 matrix is invariant under rotation.

Now, after our rotation, we have a conic with a matrix of the form

$$\begin{pmatrix} a' & 0 & \frac{d'}{2} \\ 0 & c' & \frac{e'}{2} \\ \frac{d'}{2} & \frac{e'}{2} & f' \end{pmatrix}.$$

Suppose that at least one of a' and c' is 0, so that $a'c' = 0$. Then $ac - \frac{b^2}{4} = 0$, so that $b^2 = 4ac$. Using the fact that $\theta = \frac{1}{2} \arctan\left(\frac{b}{a-c}\right)$, we can derive that $\cos 2\theta = \frac{a-c}{a+c}$ and $\sin 2\theta = \frac{b}{a+c}$. Using half-angle formulas, $\cos \theta = \sqrt{\frac{a}{a+c}}$ and $\sin \theta = \sqrt{\frac{c}{a+c}}$. Using this,

$$a' = a \cos^2 \theta + b \sin \theta \cos \theta + c \sin^2 \theta = \frac{a^2}{a+c} + \frac{b}{2} \frac{b}{a+c} + \frac{c^2}{a+c} = \frac{a^2 + 2ac + c^2}{a+c} = a+c$$

and

$$c' = a \sin^2 \theta - b \sin \theta \cos \theta + c \cos^2 \theta = \frac{ac}{a+c} - \frac{b}{2} \frac{b}{a+c} + \frac{ac}{a+c} = 0.$$

Thus, if $a'c' = 0$, then $c' = 0$. For $a+c = a' = 0$, we must have that $a = c = 0$, since $ac = \frac{b^2}{4}$ is nonnegative, so a and c have the same sign. In this case, we have also that $b = 0$, so that our conic is a line. Thus, if our conic has a nonzero quadratic term, but has $a'c' = 0$ after rotation, then $a' \neq 0$ and $c' = 0$.

Now, suppose that $a'c' \neq 0$. We translate by the vector $\left(\frac{d'}{2a'}, \frac{e'}{2c'}\right)$ to get a new conic with matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{d'}{2a'} & -\frac{e'}{2c'} & 1 \end{pmatrix} \begin{pmatrix} a' & 0 & \frac{d'}{2} \\ 0 & c' & \frac{e'}{2} \\ \frac{d'}{2} & \frac{e'}{2} & f' \end{pmatrix} \begin{pmatrix} 1 & 0 & -\frac{d'}{2a'} \\ 0 & 1 & -\frac{e'}{2c'} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a' & 0 & 0 \\ 0 & c' & 0 \\ 0 & 0 & f' - \frac{d'^2}{4a'} - \frac{e'^2}{4c'} \end{pmatrix}.$$

If the signs of a' and c' differ (so that $a'c' < 0$), this is the equation of a (possibly degenerate) hyperbola. If the signs are the same (so that $a'c' > 0$), this is the equation of a (possibly degenerate) ellipse (or an equation with no solutions).

Now, what if $a'c' = 0$? We already know that if $a' = 0$, this is a line that we can easily graph, so suppose that $a' \neq 0$. Translate by the vector $(\frac{d'}{2a'}, 0)$ to get a new conic with matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{d'}{2a'} & 0 & 1 \end{pmatrix} \begin{pmatrix} a' & 0 & \frac{d'}{2} \\ 0 & 0 & \frac{e'}{2} \\ \frac{d'}{2} & \frac{e'}{2} & f' \end{pmatrix} \begin{pmatrix} 1 & 0 & -\frac{d'}{2a'} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a' & 0 & 0 \\ 0 & 0 & \frac{e'}{2} \\ 0 & \frac{e'}{2} & f' - \frac{d'^2}{4a'} \end{pmatrix}.$$

If $e' \neq 0$, this corresponds to a parabola. If $e' = 0$, we get either a line with constant x , 2 lines with constant x , or no solutions.

Taking what we have above and compiling it, we get the following: the conic $ax^2+bx+cy^2+dx+ey+f = 0$ can be rotated to become a conic aligned with the axes. If $b^2 - 4ac = 0$, the conic is a parabola (possibly degenerate - either 1 or 2 parallel lines, or no solution). If $b^2 - 4ac > 0$, it is a hyperbola (possibly degenerate). If $b^2 - 4ac < 0$, it is an ellipse (possibly degenerate or no solutions).